

Rayan Hassan

647-975-9554 | rayan.homar5@gmail.com | linkedin.com/in/rayan-hassan-b571b7245 | github.com/CynicalD

EDUCATION

Carleton University

Bachelor of Science Honours in Computer Science

Ottawa, ON

Expected Graduation: 2028 (Third Year)

RELATED COURSES

Data Structures and Algorithms — Intro to Software Engineering — Discrete Math I, II — Operating Systems — Database Management Systems — Object-Oriented Software Engineering

PROJECTS

RAGnarok — Serverless AI Discord Assistant

Python, AWS Lambda & SAM, FastAPI, Pinecone, OpenAI, Docker

Spring/Summer 2026

- Built and deployed a serverless Retrieval-Augmented Generation (RAG) system on **AWS Lambda** that answers domain questions via a Discord `/ask` slash command, provisioned end-to-end as Infrastructure-as-Code with **AWS SAM** (Lambda, API Gateway, IAM).
- Designed a two-Lambda asynchronous architecture to satisfy Discord's 3-second response deadline: a lightweight handler verifies each request and returns a deferred acknowledgment, then invokes a worker Lambda that runs the full pipeline and delivers the answer via webhook.
- Implemented hybrid semantic + keyword retrieval over a **Pinecone** vector database, blending 1536-dimensional **OpenAI** embeddings with BM25 sparse vectors to improve recall on rare, domain-specific terminology.
- Secured the public HTTP endpoint with **Ed25519** request-signature verification (PyNaCl) and tuned cost to a fraction of a cent per query, staying within AWS and Pinecone free tiers.
- Built an offline ingestion pipeline (requests, BeautifulSoup, pandas) that scrapes, cleans, chunks (~500 tokens, 50-token overlap), and embeds wiki content into the vector index.

Cloud-Integrated File Uploader

Java, Spring Boot, React, PostgreSQL, AWS S3, Docker

Spring 2026

- Built a full-stack image-hosting web app with a React/Vite frontend and a Spring Boot REST API backed by PostgreSQL.
- Integrated Amazon S3 via the AWS SDK for efficient binary object storage of user files.
- Containerized the application using Docker to ensure consistent runtime environments across development and production.

Hintonville Farmers Market Management System

C++, QT Creator, Virtual Machine, Git & Github

Winter 2026

- Collaborated on a desktop management application allowing users to interact with market calendar, book/manage stalls, and manage system as admin.
- Designed an intuitive graphical user interface utilizing Qt Creator to handle user inputs and system navigation
- Implemented core business logic and object-oriented models in C++ to manage vendor profiles and booking states.

Health & Fitness Club Management System

Python, PostgreSQL, SQL, pycpg2, CLI

Fall 2025

- Developed a command-line Gym Management System utilizing Python and PostgreSQL to manage operations across Member, Trainer, and Admin roles
- Designed a relational database schema using SQL DDL, incorporating primary/foreign key constraints, and views to ensure data integrity across users and fitness classes.
- Integrated the pycpg2 adapter to execute dynamic SQL queries, handling user inputs for class registrations while enforcing capacity limits and preventing duplicate enrollments.

TECHNICAL SKILLS

Languages: Python, Java, C++, Javascript, SQL, HTML

Frameworks & Libraries: FastAPI, Spring Boot, React, Node.js, Express, Qt

Cloud & Dev Ops: AWS (Lambda, API Gateway, S3, IAM), AWS SAM, Docker, Git/GitHub, CI/CD

Data & AI: Pinecone (vector DB), OpenAI API, RAG, REST APIs, PostgreSQL